

Practical Feed and Speed Selection (Shop-Floor Method)

Milling Supervisor Notes

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Original Context

You are selecting feeds and speeds for milling operations based on material, tool, and operation, with practical adjustments from shop-floor experience.

Step 1: Select Surface Speed (SFM)

Material	Carbide SFM Range
Aluminum (6061)	800–1200
Aluminum (cast)	600–900
Mild Steel (1018)	300–500
Alloy Steel (4140)	200–400
Stainless Steel (304)	150–300
Tool Steel (D2)	100–250
Cast Iron	400–700
Brass	500–1000
Titanium (Ti-6Al-4V)	80–200
Nickel Alloys (Inconel)	50–150

Adjustments:

- Poor rigidity: reduce 20–40%
- Good coolant: increase slightly
- Worn machine: reduce

Step 2: Convert to RPM

$$RPM = \frac{SFM \times 3.82}{D}$$

SI equivalent:

$$N = \frac{1000 \cdot V}{\pi D}$$

where V is cutting speed in m/min and D in mm.

Step 3: Chip Load Selection

Definition:

- Small tools: diameter < 12 mm (0.5 in)

- Large tools: diameter ≥ 25 mm (1 in)

Tool Type	Chip Load (in/tooth)	Chip Load (mm/tooth)
Small endmill	0.001–0.003	0.025–0.075
Medium endmill	0.002–0.006	0.05–0.15
Large face mill	0.004–0.012	0.10–0.30

Shop rule:

If chips are dust $\rightarrow$ feed too low. If machine screams $\rightarrow$ feed too high.

Step 4: Feed Rate

$$Feed = RPM \times Z \times f_z$$

SI equivalent:

$$F = N \cdot Z \cdot f_z$$

where F is mm/min and f_z in mm/tooth.

Step 5: Practical Adjustments

Too Aggressive

- Vibration or chatter
- Insert damage
- Loud cutting noise

Action: Reduce feed first, then depth.

Too Conservative

- Powder-like chips
- Rapid tool wear
- Work hardening

Action: Increase feed.

Special Face Milling Notes

- Full-width cuts: reduce feed 20–30%
- Interrupted cuts: reduce speed 10–20%
- Long tool overhang: reduce both speed and feed

Key Insight

Feed controls chip formation and heat removal, while speed mainly affects temperature.

Supervisor Rule

Tables get you close. Chips tell you the truth.

Peck Drilling

Peck drilling is used to break chips, reduce heat, and prevent tool breakage in deep holes.

Basic Rule

Peck depth (Q) depends on diameter D :

- Shallow holes ($< 2D$): no peck needed
- Moderate depth ($2D - 5D$): $Q = 0.5D$ to $1D$
- Deep holes ($> 5D$): $Q = 0.25D$ to $0.5D$

Typical formula:

$$N = \frac{1000V}{\pi D}, \quad F = Nf$$

where f is feed per rev in mm/rev (SI).

Example

Drill: 10 mm HSS drill in mild steel

- Cutting speed: $V = 25$ m/min
- RPM: $N = \frac{1000 \cdot 25}{\pi \cdot 10} \approx 795$
- Feed per rev: $f = 0.15$ mm/rev
- Feed rate: $F \approx 120$ mm/min
- Peck depth: $Q = 5$ mm ($0.5D$)

Guidelines:

- Reduce peck as hole deepens
- Use coolant or air blast for chip evacuation
- Retract fully every 3–5 pecks in deep holes

Entry Methods for Pocket Machining

Entry strategy prevents tool damage and improves tool life.

1. Plunge Entry

- Direct vertical entry
- Use only for center-cutting tools
- Feed: 30–50% of normal cutting feed

Example: 12 mm endmill, normal feed = 800 mm/min

- Plunge feed: 300–400 mm/min

2. Ramp Entry

- Tool enters at angle (1–3 degrees typical)
- Safer than plunge
- Use for most pocketing operations

Ramp angle relation:

$$\tan(\theta) = \frac{\text{depth}}{\text{horizontal travel}}$$

Example: Depth = 10 mm, angle = 2°

- Horizontal distance ≈ 286 mm

3. Helical Entry

- Spirals into material
- Best load distribution
- Preferred for deep pockets

Typical parameters:

- Helix diameter: 1.2–1.5× tool diameter
- Pitch: 0.5–1.0 mm per revolution

Example: 16 mm endmill

- Helix dia: 20–24 mm
- Pitch: 0.8 mm/rev

4. Pre-drill Entry

- Drill hole first, then mill
- Useful for hard materials or large depths

Supervisor Insight

Never force a tool straight into material unless it is designed for it. Entry strategy determines tool life more than cutting speed.